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Postdoctoral Fellow @ Georgia Institute of Technology

Employment

- 2025-Now **Postdoctoral Fellow**, [Hybrid Intelligence \(HI\) Lab](#), Georgia Institute of Technology
(PI: [Asst Prof. Mengyao Li](#), and [Prof. Jamie Gorman](#))
- 2026 Spr **Adjunct Course Instructor**, Georgia Institute of Technology
- 2025 **Research Fellow** (Human Factors), [Air Traffic Management Research Institute \(ATMRI\)](#), Nanyang Technological University
(PI: [Assoc Prof. Mir Feroskhan](#))
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Education

- 2022-25 **Ph.D.**, Hong Kong Polytechnic University
- 2020-21 **M.Sc.**, Nanyang Technological University
- 2013-17 **B.Eng.**, Dalian University of Technology
 B.Art (Dual)
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Publications

 [Google Scholar](#)

* → Corresponding author

† → Equal contribution

Journal Articles

- [J1] Y. Sun, D. Li, S. Han, **Mengtao, Lyu**, and F. Li*, “Arhud design for dynamic spatial information presentation to improve driver situation awareness of blind spots,” *International Journal of Human-Computer Studies*, p. 103 793, 2026.
- [J2] Z. Li, F. Li*, and **Mengtao, Lyu**, “Tracking the unseen and unaware: Deciphering controllers’ detection failures to warnings through eye-tracking metrics,” *International Journal of Human-Computer Interaction*, pp. 1–20, 2025.
- [J3] **Mengtao, Lyu** and F. Li*, “Do you need help? identifying and responding to pilots’ troubleshooting through eye-tracking and large language model,” *International Journal of Human-Computer Studies*, p. 103 617, 2025.

- [J4] **Mengtao, Lyu**, F. Li*, C.-H. Lee, and C.-H. Chen, "Valio: Visual attention-based linear temporal logic method for explainable out-of-the-loop identification," *Knowledge-Based Systems*, vol. 299, p. 112 086, 2024.
- [J5] **Mengtao, Lyu**, F. Li*, X. Qu, and Q. Li, "Flashlight model: Integrating attention distribution and attention resources for pilots' visual behaviour analysis and performance prediction," *International Journal of Industrial Ergonomics*, vol. 103, p. 103 630, 2024.
- [J6] Q. Li, K. K. Ng*, C. Simon, C. Y. Yiu, and **Mengtao, Lyu**, "Recognising situation awareness associated with different workloads using eeg and eye-tracking features in air traffic control tasks," *Knowledge-Based Systems*, vol. 260, p. 110 179, 2023.
- [J7] **Mengtao, Lyu**, L. Fan*, X. Gangyan, and H. Su, "Leveraging eye-tracking technologies to promote aviation safety-a review of key aspects, challenges, and future perspectives," *Safety science*, vol. 168, p. 106 295, 2023.
- [J8] **Mengtao, Lyu**, X. Li*, and C.-H. Chen, "Achieving knowledge-as-a-service in iiot-driven smart manufacturing: A crowdsourcing-based continuous enrichment method for industrial knowledge graph," *Advanced Engineering Informatics*, vol. 51, p. 101 494, 2022.
- [J9] X. Li*, **Mengtao, Lyu**, Z. Wang, C.-H. Chen, and P. Zheng, "Exploiting knowledge graphs in industrial products and services: A survey of key aspects, challenges, and future perspectives," *Computers in Industry*, vol. 129, p. 103 449, 2021.
- [J10] Y. Deng[†], **Mengtao, Lyu**[†], J. Gao, J. Xiao, and M. Feroskhan*, "Trust-up: Trustworthy reinforcement learning using safe techniques for uav pursuit," *Advanced Engineering Informatics*, *Accepted*, 2026.

Conference Proceedings

- [C1] Y. Sun, D. Dillion, K. Gray, **Mengtao, Lyu**, Z. Zhang, and F. Li, "From expectation to evaluation: Expectation cues systematically bias llm and human judgment," in *CHI '26: Proceedings of the 2026 CHI Conference on Human Factors in Computing System*, Association for Computing Machinery, 2026.
- [C2] **Mengtao, Lyu** and F. Li, "Patterning risk: An innovative task design method for simulating incidents in transportation studies," in *Advances in Human Factors of Transportation*, vol. 148, AHFE Open Acces, 2024.
- [C3] **Mengtao, Lyu**, F. Li, and C.-H. Lee, "The effects of adaptive automation on pilots' flight control performance and visual attention distribution," in *Leveraging Transdisciplinary Engineering in a Changing and Connected World*, IOS Press, 2023, pp. 82–91.
- [C4] Z. Li, M.-H. Hsieh, M. P. Ramesh, I. Dhief, **Mengtao, Lyu**, and M. Feroskhan, "How multi-sector planning shapes shared situation awareness among air traffic controllers: A team-level eye-tracking evaluation," in *28th International Conference on Human-Computer Interaction (HCII 2026)*, Springer Nature Switzerland, 2026, *Accepted*.
- [C5] **Mengtao, Lyu** and Q. Li, "Methodological challenges in human factors research for air traffic management," in *2026 IEEE International Conference on Human-Machine Systems (ICHMS)*, 2026, *Accepted*.

- [C6] **Mengtao, Lyu**, E. Rojas, M. Li, B. Bracken, N. Cooke, P. Desrochers, J. Gorman, L. Huang, M. Kilcullen, M. Rosen, M. Scalia, G. A. Yadav, X. Yin, E. Zahmat Doost, and S. Zhou, “Beyond dyads: Machine learning for team performance prediction in multi-member teams,” in *ASPIRE International Annual Meeting (HFES 2026)*, SAGE Publications, 2026, Accepted.
- [C7] M. P. Ramesh, **Mengtao, Lyu**, I. Dhief, Z. Li, and M. Feroskhan, “Human-ai decision support for sustainable atco workload: Forecast-informed, bilevel-optimized task-load estimation,” in *28th International Conference on Human-Computer Interaction (HCII 2026)*, Springer Nature Switzerland, 2026, Accepted.
- [C8] E. Rojas, **Mengtao, Lyu**, M. Li, B. Bracken, N. Cooke, P. Desrochers, J. Gorman, L. Huang, M. Kilcullen, M. Rosen, M. Scalia, G. A. Yadav, X. Yin, E. Zahmat Doost, and S. Zhou, “Identifying latent processes in multi-modal physiological signals for team performance: An exploratory approach,” in *ASPIRE International Annual Meeting (HFES 2026)*, Under Review, SAGE Publications, 2026, Accepted.

Working papers

- [W1] Q. Li, X. Yuan, K. K. Ng, and **Mengtao, Lyu***, *Translating gaze-based human reliability evidence into cockpit safety functions: A transparent pathway from dual-pilot cockpit hindrance to single-pilot flight deck evolution*, Reliability Engineering System Safety, under review, 2026.
- [W2] Z. Li, **Mengtao, Lyu**, F. Li*, H. Qiao, T. Yang, and C.-H. Chen, *Predicting detection failures in air traffic control: Leveraging fixation-state transitions and llm-driven probabilistic threshold tuning*, Advanced Engineering Informatics, under review, 2026.
- [W3] **Mengtao, Lyu**, M. Li, and Q. Li*, *The promise and peril of generative ai for hci in autonomous driving: A critical survey*, International Journal of Human-Computer Interaction, Working In Progress, 2026.
- [W4] **Mengtao, Lyu**, E. Rojas, M. Li, X. Yin, E. Zahmat Doost, B. Bracken, N. Cooke, P. Desrochers, L. Huang, M. Kilcullen, M. Rosen, M. Scalia, W. Aaron, G. A. Yadav, S. Zhou, and J. Gorman, *Predicting team performance with multimodal data using a dual attention framework and identifying critical psychophysiological features*, IEEE Transactions on Pattern Analysis and Machine Intelligence, Under Review, 2026.

Talks & Presentations

- [T1] *Enhancing pilot performance through eye-tracking and ai: A closed-loop approach to aviation safety*, The Hong Kong Ergonomics Society (HKES), May 2026.
- [T2] *Interactive robotics and human-centered design*, Center for Human-AI-Robot Teaming (CHART), Georgia Tech, May 2026.
- [T3] *Learning as a system: Brain, body, and design*, Faculty Panel and Lunch, Georgia Tech, Mar. 2026.
- [T4] *Through the lens of situation awareness: The promise and peril of generative ai for hci in autonomous driving*, HFES HART TG Monthly Member Conversation, Feb. 2026.
- [T5] *Panel talk*, HFES- Georgia Tech Student Chapter, Oct. 2025.

Academic Service

Journal/Conference Reviewing

Journal

IEEE Transactions on Human-Machine Systems, International Journal of Human-Computer Interaction, Ergonomics, Engineering Applications of Artificial Intelligence, Advanced Engineering Informatics, Knowledge-based systems, Computers & Industrial Engineering, Frontiers in Neuroscience

Conference

ACM International Conference on Multimedia (MM), ASPIRE International Annual Meeting (HFES), International Conference on Human-Computer Interaction (HCII), IEEE International Conference on Human-Machine Systems (ICHMS), IEEE International Conference on Cyberworlds (CW), International Conference on Transdisciplinary Engineering (TE)

Selected Awards & Honors

2025	Inspiring, innovative and pioneering research	Tobii
2025	Outstanding Student Project Award	The Hong Kong Ergonomics Society (HKES)
2025	IEA/Kingfar Student Award	The International Ergonomics Association (IEA)
2022	Postgraduate Studentship Award	Hong Kong Polytechnic University

Projects

2025-26, **Defense Advanced Research Projects Agency (DARPA)**
Bio-Behavioral Team Dynamics Measurement System (BioTDMS)

2025, **International Centre for Aviation Innovation (ICAI) Singapore**
Multi Sector Planning

2025, **University Grants Committee (UGC)**, Early Career Scheme (ECS)
Looking but Not Seeing? Mitigating Inattentional Blindness in Remote Digital Towers for Safe and Watchful Air Traffic Control

2025, **National Natural Science Foundation of China (NSFC)**, Young Scientists Fund
Research on the Degradation Mechanism of Pilot Vigilance in Response to Dynamic Task Workload Conditions

2025, **COMAC**
AI-driven Virtual Co-Pilots in Aviation

2023-25, **Hong Kong Productivity Council (HKPC)**, Smart Traffic Fund (STF)
Driving Style-based Adaptive Virtual Training Platform: Build Safe Human Driving Habits in Autonomous driving

2023-25, **Hong Kong Polytechnic University**
Investigating pilots' decision-making and cognitive performance in response to acute stress

2023-24, **Hong Kong Polytechnic University**
Improved Student Attention and Engagement via Visual Attention Analytics-based Slides and Lesson Plan Design

2022-24, **Hong Kong Polytechnic University**
Visual Attention Analytics in Aviation Automation via Deep Learning Model

2021-23, **Hong Kong Polytechnic University**
Psychophysiological Data-based Knowledge Transfer Model for Adaptive Training

2020-21, **National Research Foundation (NRF) Singapore**
Stakeholder-oriented innovative user-centric design and solution generation

Teaching

Georgia Institute of Technology

2026 **Instructor**, Science of Occupational Health and Safety (COS2801)

Hong Kong Polytechnic University

2023 **Teaching Assistant**, Human Factors, Accident Prevention and Aircraft Maintenance (AAE5002)

2023 **Teaching Assistant**, Human Factors in Aviation (AAE4903)

2022 **Teaching Assistant**, Safety, Reliability and Compliance (AAE3006)

Advising

Graduate

2025-	Mercedes P. Ramesh	Nanyang Technological University	PhD Thesis
2023-	Yiteng Sun	Hong Kong Polytechnic University	PhD Thesis
2023-24	Donglin LI	Hong Kong Polytechnic University	Master Dissertation
2023-24	Henry CHAN	Hong Kong Polytechnic University	Master Dissertation

Undergraduate

2023-24	Cheung, Choi, & Kwok	Hong Kong Polytechnic University	Capstone Project (FYP)
2022-23	Leung, Tsang, & Lam	Hong Kong Polytechnic University	Capstone Project (FYP)

Other Experience

Industrial

2019-20	IT Supervisor , Asics China Trading Co., Ltd.
2017-19	SAP Consultant , ABeam Global Development Centre (Shanghai) Co., Ltd.

Certificates

2022	Adult Cardio Pulmonary Resuscitation (ACPR)
2016	N1, Japanese-Language Proficiency Test (JLPT)

Last updated: June 2, 2026